6.1 Exponential Growth and Decay Functions

Name _____ Period ____ Date _____

Algebra II 6.1 · Student sheet

I can…

- identify and evaluate exponential functions.
- tell exponential growth from decay and find the percent rate of change.
- write and use exponential models, including compound interest.

Vocabulary · fill in the missing word

Exponential function — $y =$ _____, where $a \neq 0$, $b > 0$, and $b \neq 1$

Exponential growth / decay — _____

Asymptote — a line a graph gets _____ and closer to; for $y = ab^x$ it is $y = 0$

Growth / decay model — $y = a(1 + r)^t$ / $y = a(1 - r)^t$, where r is the _____ rate as a decimal

Compound interest — $A =$ _____

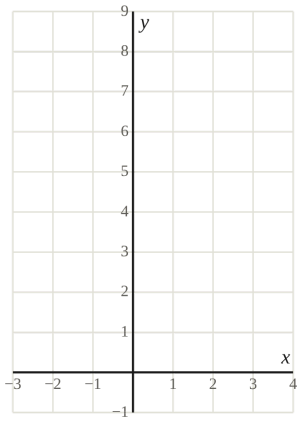
Part 1 · Paper folding

Fold a sheet of paper in half again and again. After each fold, record the number of layers and the fraction of the original sheet showing on top.

Folds x	0	1	2	3	4	5
Layers	1					
Fraction on top	1					

Equation for the layers: $y =$ _____ Equation for the fraction: $y =$ _____

Graph both equations for $x = -2$ to 3 on the same grid.



How are the two graphs alike? How are they different?

Part 2 · Notes: fill in the blanks

An **exponential function** has the form $y =$ _____, where $a \neq 0$, $b > 0$, and $b \neq 1$.

If $b > 1$, the function shows exponential _____. If $0 < b < 1$, it shows exponential _____.

The graph of $y = ab^x$ has a horizontal asymptote at _____.

Growth: $y = a(1 + r)^t$. Decay: $y =$ _____.

Compound interest: $A =$ _____

Part 3 · Examples (work them with the class)

Example 1

Evaluate $y = 4(3)^x$ for $x = -2$ and $x = 2$.

Example 2

Determine whether each function represents exponential growth or exponential decay.

a. $y = 2(0.6)^x$ b. $y = 0.5(1.25)^x$

Example 3

The amount y (in milligrams) of a medicine in your bloodstream t hours after a dose is $y = 200(0.85)^t$. Is it growth or decay? Find the percent rate of change. How much is left after 3 hours?

Example 4

Rewrite $y = a(1.02)^{12t}$ in the form $y = a(1 + r)^t$. Find the yearly percent rate of change.

Example 5

You deposit $5,000 in an account that pays 4% annual interest compounded monthly. Find the balance after 3 years.

Example 6

A car is worth $18,500. Its value decreases by 12% each year. Write a model and estimate its value after 6 years.

Part 4 · Partner practice

Partner 1

The number of users of a website t years after it launched is $y = 1500(1.03)^t$. Is it growth or decay? Find the percent rate of change.

Partner 2

You deposit $2,000 in an account that pays 6% annual interest compounded quarterly. Find the balance after 5 years.